

Caps and Floors

5.1 A cap is a strip of options, not one option

An **interest rate cap** pays its holder the excess, if any, of a floating rate over a fixed **strike** or **cap rate** K , on each of a series of accrual periods. Critically, a cap is not a single option: it is a portfolio of independent options, one per accrual period, each called a **caplet**.

Key result: the caplet payoff

For a forward-looking rate over the period $[T_1, T_2]$ with accrual fraction τ , notional N , and strike K , the caplet pays, at the payment date (conventionally T_2 , or occasionally T_1 depending on the product):

$$\text{Caplet payoff} = N\tau \max(L(T_1; T_1, T_2) - K, 0)$$

A **floorlet** pays the mirror-image $N\tau \max(K - L(T_1; T_1, T_2), 0)$. A **cap** is the sum of caplets across all periods; a **floor** is the sum of floorlets. Because each caplet's payoff depends only on the rate fixing for its own period, caplets can — and in practice must — be priced and risk-managed individually, even though they trade packaged together as a single cap.

PITFALL

The diffusion that determines the option's uncertainty runs to the *fixing* date T_1 — that is where the randomness of $L(T_1; T_1, T_2)$ is resolved. The discounting, however, runs to the *payment* date, typically T_2 . Conflating the two — using T_1 in the discount factor, or using T_2 in the volatility term $\sqrt{T}$ — is a common and easy mistake. The mnemonic worth keeping: *uncertainty stops at the fixing; cash arrives at the payment*.

5.2 Cap–floor parity

Exactly as put–call parity constrains equity option prices, an analogous identity constrains caps and floors, and it is worth deriving because it gives a fast internal consistency check on any pricing engine.

Key result: cap–floor parity

For a caplet and floorlet on the same rate, same strike, same period:

$$\text{Caplet} - \text{Floorlet} = N\tau P(0, T_2) (F - K)$$

where $F = L(0; T_1, T_2)$ is today's forward rate for the period. This follows immediately from $\max(F - K, 0) - \max(K - F, 0) = F - K$ applied under the appropriate discounting, taking expectations under the forward measure (Chapter 12 makes this rigorous). Summed across all periods of a cap and floor with the same strike, this becomes the statement that a cap minus a floor equals a receive-fixed swap struck at K ,

in present-value terms.

This identity is a genuine, cheap sanity check worth building directly into a pricing or calibration system: if a cap and floor at the same strike, priced independently through two separate code paths, don't satisfy parity to within numerical tolerance, something in the pipeline is broken — long before any volatility-model subtlety becomes relevant.

5.3 The in-arrears caplet, revisited

Chapter 3 flagged that the dominant SOFR cap convention today is backward-looking, compounded-in-arrears, not the forward-looking payoff above. It is worth stating the in-arrears payoff explicitly, because it is the version that actually trades.

Key result: the in-arrears caplet payoff

Writing $R_{\text{cmp}}(T_1, T_2)$ for the rate realised by compounding the overnight fixings across $[T_1, T_2]$ (as defined in Chapter 3), the in-arrears caplet pays, at T_2 :

$$N\tau \max(R_{\text{cmp}}(T_1, T_2) - K, 0)$$

This looks identical in form to the forward-looking caplet, but $R_{\text{cmp}}(T_1, T_2)$ is only fully known at T_2 , having accumulated information throughout $[T_1, T_2]$ rather than being fixed once at T_1 . The payoff is therefore genuinely path-dependent within the period, and a model that treats $R_{\text{cmp}}(T_1, T_2)$ as if it behaved like a single lognormal or normal draw fixed at T_1 — the classical LIBOR-caplet assumption — introduces a small but systematic **in-arrears convexity** bias, as Worked Example 3.1 showed.

Worked example 5.1 — pricing a full cap as a strip.

A 2-year SOFR cap has quarterly caplets on notional \$10mm, strike $K = 4.00\%$, with forward rates for the four periods (annualised, already adjusted for the relevant convention) of 3.80%, 3.95%, 4.10%, 4.20%, accrual fractions $\tau_i = 0.25$, and discount factors to each payment date $P(0, T_i) = 0.9901, 0.9803, 0.9702, 0.9598$. Suppose (for this illustration only — Chapter 6 builds the option pricing formula properly) that each caplet's intrinsic-forward value is used directly, i.e. the simplification $\max(F_i - K, 0)$ evaluated at the forward, with no time value. Compute the resulting undiscounted-then-discounted total.

Step 1 — intrinsic value at each period.

$$\begin{aligned} \max(3.80\% - 4.00\%, 0) &= 0, & \max(3.95\% - 4.00\%, 0) &= 0, \\ \max(4.10\% - 4.00\%, 0) &= 0.10\%, & \max(4.20\% - 4.00\%, 0) &= 0.20\% \end{aligned}$$

Step 2 — cash flow per period.

$$\text{CF}_3 = \$10\text{mm} \times 0.25 \times 0.0010 = \$2,500, \quad \text{CF}_4 = \$10\text{mm} \times 0.25 \times 0.0020 = \$5,000$$

Step 3 — discount and sum.

$$V \approx \$2,500 \times 0.9702 + \$5,000 \times 0.9598 = \$2,426 + \$4,799 = \$7,225$$

Sanity check. Only the two periods where the forward rate exceeds the strike contribute, exactly as expected for a call-type payoff; the first two periods, being out-of-the-money at the forward, contribute nothing under this simplified intrinsic-only treatment.

Interpretation. This number is a *lower bound*, not the real cap price — it ignores time value entirely, pricing each caplet as if the forward rate were certain to realise exactly at its current forward level. The true value is higher, because there is a positive probability that even the currently-out-of-the-money first two caplets finish in-the-money, and volatility is worth something. Chapter 6 replaces this intrinsic-only shortcut with the real Black and Bachelier formulas and recomputes this exact cap properly.

ON THE DESK

Caps and floors are the standard building block of floating-rate loan hedging: a corporate borrower with SOFR-linked debt buys a cap to put a ceiling on their borrowing cost, paying an upfront premium in exchange for protection. This is a genuinely different economic use case from a swaption, which is typically about hedging or expressing a view on the *level of rates at a single future decision point* (should I enter this swap or not), whereas a cap is about capping a *stream* of floating exposure across many periods. Both instruments are priced with the same Black/Bachelier machinery (Chapter 6), but a desk's cap book and swaption book are usually risk-managed somewhat separately because their underlying exposures — a strip of independent forward rates versus a single forward swap rate — have different correlation structure.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. Write the caplet and floorlet payoffs and explain why a cap is priced as a strip of independent options rather than one option.
2. State and use cap–floor parity as a sanity check.
3. Explain why an in-arrears (compounded, backward-looking) caplet is not simply a relabelled forward-looking caplet, and what kind of bias results from treating it as one.

